SCHUR-IMAGE BASES OF $G_{(2^k)}$: POSITIVE FOR $k \le 5$ BY CHOICE, IMPOSSIBLE AT $k = 6, 7$ (ANSWER TO CLIO'S 469 QUESTION)

RICK

From: Rick. To: Clio Vega (cc Robin Langer). Date: 2026-11-14.
WIP commit: `d8b252a`
Scripts and outputs: `scripts/scope296/q75-positivity/` (this repository).

Grades. Every claim in this note is graded `computed`: exact arithmetic over $\mathbb{Q}$ (integer Littlewood–Richardson coefficients, `Fraction` linear algebra), $k \le 7$ only. Nothing here is `proved` for general k. The one cited fact (that $G_{(2,2)} \cong H^*\mathrm{Gr}(2,4)$ as rings) comes from the 2026-11-12 Q75 note.

Your question (UID 469). Does $G_{(2^k)}$ have a basis with nonnegative structure constants, for example the images of the s_λ? **Answer.** For Schur-subset bases the answer is yes for $k \le 5$, but only for some choices of subset, and no for $k = 6$ and $k = 7$, by exhaustive search.

1. Setup

Let $R = r_{(2^k)}$ be the ribbon Schur function with k rows of length 2, so $d = \deg R = 2k$. Let $G = G_{(2^k)} = \mathrm{Sym}_{\mathbb{Q}}/\operatorname{Ann}(R)$, where f acts by $f^\perp$ for the Hall pairing. The image of f in G_i is the vector $\big(\langle f s_\nu, R\rangle\big)_{\nu \vdash 2k-i}$. A *Schur-subset basis* picks, in each degree n, a set B_n of $r_n = \dim G_n$ partitions of n whose images are linearly independent. Its structure constants $c^\nu_{\lambda\mu} \in \mathbb{Q}$ are defined by $s_\lambda s_\mu = \sum_{\nu \in B_{|\lambda|+|\mu|}} c^\nu_{\lambda\mu} s_\nu$ in G.

2. Shape of G

Hilbert function. The computed values are $1,1,2,1,1$ ($k = 2$); $1,1,2,2,2,1,1$ ($k = 3$); $1,1,2,2,3,2,2,1,1$ ($k = 4$); $1,1,2,2,3,3,3,2,2,1,1$ ($k = 5$), and they continue the same way for $k = 6, 7$. Each equals $\begin{bmatrix} k+2 \\ 2 \end{bmatrix}_q$, as the Q75 theorem says.

Schur images are not the box. The nonzero s_λ images are linearly dependent, and the set of λ with $s_\lambda \ne 0$ is not the $2 \times k$ box. At $k = 2$: in degree 3, s_3 and s_{21} are both nonzero and span a line; in degree 4, $s_{31} \ne 0$ and $s_{22} \ne 0$, while $s_4 = s_{211} = s_{1111} = 0$. So the isomorphism $G_{(2,2)} \cong H^*\mathrm{Gr}(2,4)$ does not send s_λ to s_λ, because in $H^*\mathrm{Gr}(2,4)$ both s_3 and s_{31} vanish.

Number of Schur-subset choices of B_n, for $n = 0, \dots, 2k$.

k	$\#B_n$
2	1, 1, 1, 2, 2 (total 4)
3	1, 1, 1, 2, 5, 4, 5 (total 200)
4	1, 1, 1, 2, 5, 13, 24, 9, 11 (total 308 880)
5	1, 1, 1, 2, 5, 13, 94, 59, 105, 19, 23
6	1, 1, 1, 2, 5, 13, 94, 292, 783, 257, 426, 37, 46
7	1, 1, 1, 2, 5, 13, 94, 292, 4472, 2526, 6187, 947, 1562, 70, 86

These counts do not depend on any order, but the basis that a greedy procedure returns does. Positivity depends on that choice, as the next section shows.

3. Positivity, by exhaustive search

- $k = 2$: all 4/4 Schur-subset bases have nonnegative constants.
- $k = 3$: 160/200 are positive.
- $k = 4$: 19 008/308 880 are positive. The lex-first greedy basis (row-heavy first) fails: $s_1 s_{31}$ has coefficient -3 on s_5. The lex-last greedy basis (column-heavy first) also fails: $s_1 s_{22}$ has coefficient $-5/12$ on s_{2111}.

Date: 2026-11-14.

- $k = 5$: at least 10^6 positive bases (the search stops at 10^6). One of them, $B_6 = \{s_6, s_{51}, s_{33}\}$, $B_7 = \{s_{61}, s_{43}\}$, $B_8 = \{s_{62}, s_{44}\}$, ..., was re-verified separately. All its constants are ≥ 0 and their denominators are $1, 2, 3, 4, 6, 12$.
- *Integrality.* If we ask for integral as well as positive constants (unscaled s_λ), $k = 2$ has 2 such bases and $k = 3, 4, 5$ have none.

$k = 6, 7$: no positive Schur-subset basis exists. The search is exhaustive and dies in degree 8. At $k = 6$ there are exactly 128 positive partial bases $B_0, \ldots, B_7$, using only 4 distinct choices of B_7, and none of them extends to degree 8. Every one of the 128 contains $s_1,\ s_2,\ s_{11},\ s_{33},\ s_{22111}$. *Certificate:* the three products

$$s_{11}s_{33}, \qquad s_1 s_{22111}, \qquad s_2 s_{33}$$

cover all 783 choices of B_8: for each B_8, at least one of the three products has a negative coefficient. The first product is the witness for 162 choices, the second for 618, and the third for 3. The certificate script was re-run for this note and its output matches the stored one exactly. At $k = 7$ the result is as recorded: *NO (exhaustive; dies in degree 8).* The per-degree counts of surviving B_n are $1, 1, 1, 2, 4, 4, 12, 4, 0, \ldots, 0$, the same profile as at $k = 6$. No separate $k = 7$ certificate has been extracted.

4. Controls and alternatives

Controls. (C1) The LR engine passes the dimension identity $\sum_\nu c^\nu_{\lambda\mu} f^\nu = \binom{n}{|\lambda|} f^\lambda f^\mu$ on all 301 products with $n \leq 8$, with no failures. (C2) Running the same pipeline with $R = s_{(k,k)}$ reproduces $H^*\mathrm{Gr}(2, k+2)$ for $k = 2, \ldots, 5$: the nonzero s_λ are exactly the $2 \times k$ box, they are independent, and every structure constant equals the LR coefficient ($12, 52, 153, 363$ constants, none negative).

Alternative bases. (i) *Gram-dual basis.* Take the basis Poincaré-dual to a Schur basis. It already has a negative constant at $k = 2$: the degree-2 Gram matrix on $\{s_2, s_{11}\}$ is $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, and the product of the two dual elements is -1 times the top class. (ii) *Cone-edge basis.* Take one Schur image on each extreme ray of the cone of Schur images in G_n. For $k \leq 4$ this cone is simplicial in every degree. Every s_λ is then ≥ 0 in the edge basis, so all structure constants are ≥ 0. The construction breaks at $k = 5$, degree 6, where the cone has 5 extreme rays in dimension 3. At $k = 6$ the cone is non-simplicial in degrees $6, 7, 8$.

5. Conclusion (hedged)

If "deformation of LR" means nonnegative structure constants on a basis of Schur images, that reading fails for $G_{(2^k)}$ from $k = 6$ on. We checked $k = 6, 7$ exhaustively. We have not checked $k \geq 8$, and we do not claim that $k = 6, 7$ settle it. For $k \leq 5$, positive Schur-subset bases exist, but they are chosen rather than canonical, and they are not integral beyond $k = 2$. *Open:* is there a positive basis that is not made of Schur images, for example one coming from a cellular structure or a Gröbner (standard-monomial) basis of $\mathrm{Ann}(R)$?

Scripts (in `scripts/scope296/q75-positivity/`, mirrored from Rick's local `proofs/scope296/q75-positivity/`): `engine.py` (exact LR and the quotient), `survey.py`, `positivity.py` ($k \leq 4$ and greedy orders), `fastsearch.py` and `fastsearch_int.py` (exhaustive search, positive and integral), `verify_k5.py`, `obstruction_k6.py`, `cert_k6.py` (the certificate), `cone.py`, `dualbasis.py`, `control.py`. Each script's output is stored next to it as `*.txt`, and `results.txt` summarises them.